



AIMS

African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 13: Tests of Hypothesis

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- 2 Testing normal mean / unknown variance
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Testing normal mean

Example

The diameter of steel shafts produced by a certain manufacturing process should have a mean diameter of 0.255 inches.

The diameter is known to have a standard deviation of $\sigma = 0.0001$ inch.

A random sample of 10 shafts has an average diameter of 0.2545 inch.

$$(X_1, X_2, \dots, X_{10}) \quad \rightarrow \quad \bar{X}_{10} = \frac{1}{10} \sum_{i=1}^{10} X_i = 0.2545 \text{ inch.}$$

$$X_i \sim \mathcal{N}(\mu, \sigma^2), \quad \mu = 0.255 \text{ and } \sigma^2 = 10^{-8}.$$

Does the machine need to be adjusted?

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Does the machine need to be adjusted?

- $H_0 : \mu = 0.255$ versus $H_a : \mu \neq 0.255$
- Decision rule: $\delta(x) = 1$ if $|\bar{X}_n - \mu| > C$

→ If $|\bar{X}_n - 0.255| > C$, we consider that the machine needs to be adjusted.

Testing normal mean

➡ Critical region $R = \{\underline{X} \in \mathcal{X} \text{ such that } |\bar{X}_n - 0.255| > C\}$

Test statistics: $Z_n = \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}}$, $n = 10$

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$$P(\underline{X} \in R \mid \mu = 0.255) = \alpha$$

Testing normal mean

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$$\begin{aligned} P(\underline{X} \in R \mid \mu = 0.255) = \alpha &\iff P(|Z_n| > C' \mid \mu = 0.255) = \alpha \\ &\iff P(Z_n \leq -C') + P(Z_n > C') = \alpha \\ &\iff \Phi(-C') + 1 - \Phi(C') = \alpha \\ &\iff 2(1 - \Phi(C')) = \alpha \iff C' = \Phi^{-1}(1 - \alpha/2). \end{aligned}$$

➡ Hence the critical region has the form:

$$R = \{\underline{X} \in \mathcal{X} \text{ such that } |Z_n| > \Phi^{-1}(1 - \alpha/2)\}.$$

- Conclusion

- For $\alpha = 5\%$, $\Phi^{-1}(0.975) = 1.96$.

The value of the test statistic is

$$|Z_n| = \left| \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} \right| = \left| \frac{0.2545 - 0.255}{0.0001/\sqrt{10}} \right| = 15.81.$$

We are in the critical region since $\{|Z_n| > 1.96\}$.

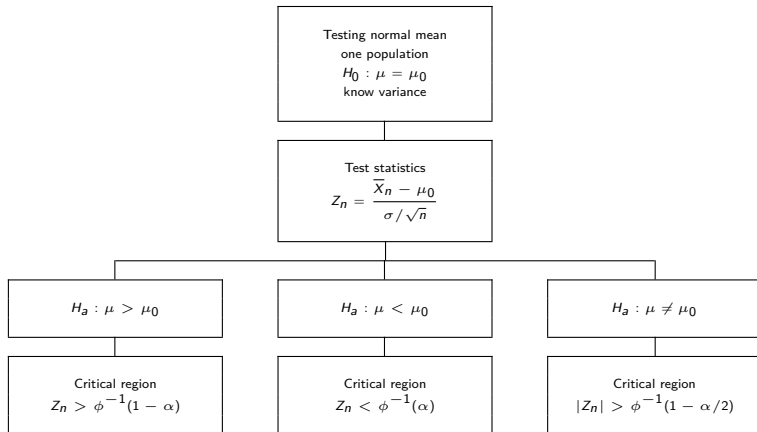
⇒ We reject the null hypothesis with probability to be wrong 5%.

→ The machine needs to be adjusted.

- Here the p -value is $P(|Z_n| > 15.81) \approx 0$

⇒ No chance to be wrong rejecting H_0 !

Testing normal mean / known variance



Testing normale mean unknown variance

When the **variance is unknown**, it can be estimated:

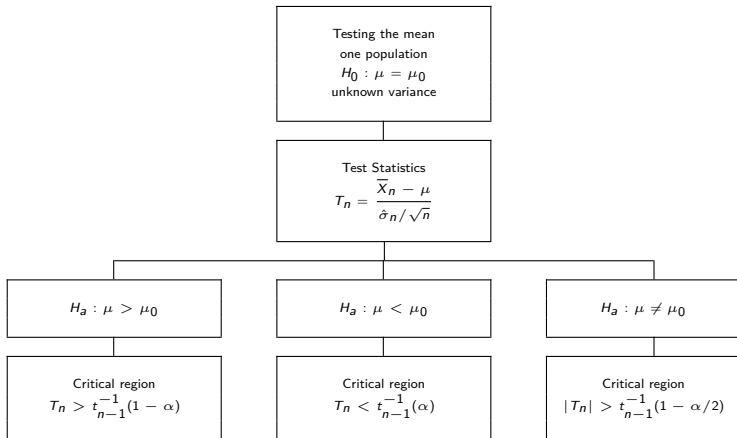
$$\hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Suggested *Test Statistics*:

$$T_n = \frac{\bar{X}_n - \mu}{\hat{\sigma}_n / \sqrt{n}}.$$

➡ T_n follows a Student distribution with $n - 1$ d.o.f.

Testing normal mean / unknown variance



Testing normal mean unknown variance

Exercise

A light-bulb manufacturer claims that the light bulbs produced by his/her company have an average life of 1200 hours.

A consumer protection agency wishes to test this claim against the alternative of fraudulent advertising (the average life is less than 1200 hrs).

A random sample of 16 bulbs is chosen from production and tested as to life times.

The summary statistics turn out to be $\bar{X}_{16} = 1148.5$ and $\hat{\sigma}_{16}^2 = 12544.224$.

What conclusions are reached?

Testing normal mean unknown variance

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What conclusions are reached?

Solution:

$H_0 : \mu = 1200$ versus $H_a : \mu < 1200$.

$n = 16$, $\alpha = 0.05$.

$$T_{16} = \frac{\bar{X}_{16} - \mu_0}{\hat{\sigma}_{16}/\sqrt{16}} = \frac{1148.5 - 1200}{\sqrt{12544.224}/\sqrt{4}} = -1.8393.$$

$$t_{15}^{-1}(.05) = -1.753.$$

We have: $T_n < t_{n-1}^{-1}(\alpha)$. Therefore, we are in the critical region.

Conclusion: We reject H_0 with a probability to be wrong 5% (p-value: 0.0428).

➡ Considering that the average lifetime is not equal to 1200, we have a 5% chance to be wrong. *High probability for the claim to be fraudulent!*

Testing normal variance

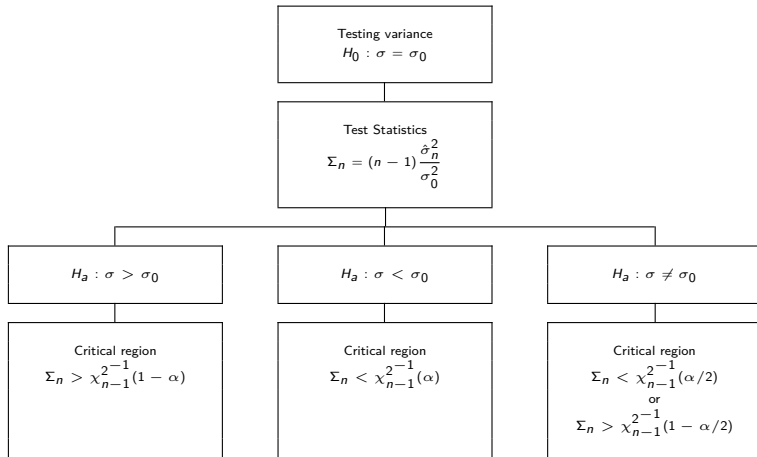
- Testing the variance of a normal distribution (one population)

$$H_0 : \sigma^2 = \sigma_0^2 \text{ versus } H_a : \sigma^2 > \sigma_0^2.$$

- ➡ Test Statistics: $\Sigma_n = (n-1)\hat{\sigma}_n^2/\sigma_0^2$
- Under H_0 , $(n-1)\hat{\sigma}_n^2/\sigma_0^2$ has a chi-square with $n-1$ d.o.f.
- Critical region: $R = \{\Sigma_n > c\}$.
- For a given α , c can be computed and the critical region is:

$$\{\Sigma_n > \chi_{n-1}^{2^{-1}}(1-\alpha)\}.$$

Testing normal variance



Testing normal variance

Exercise

The standard deviation of the diameter of parts produced by a certain machine has a specification that it should not be greater than 0.5 mm. But sometimes the standard deviation “shifts”. If the standard deviation is determined to exceed 0.5 mm, the production process is stopped and the machine is adjusted.

A random sample of 14 items is taken from the production of this machine. Diameters of these items are measured and the standard deviation is computed: 0.64 mm is found.

Should we stop the production process to adjust the machine?

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Solutions

$$H_0 : \sigma^2 = 0.25 \text{ versus } H_a : \sigma^2 > 0.25.$$

$$\Sigma_n = (n-1) \frac{\widehat{\sigma}_n^2}{\sigma_0^2} = 13 \times \frac{(0.64)^2}{0.25} = 21.2992.$$

$$\chi_{n-1}^{2-1}(1-\alpha) = \chi_{13}^{2-1}(0.95) = 22.3621.$$

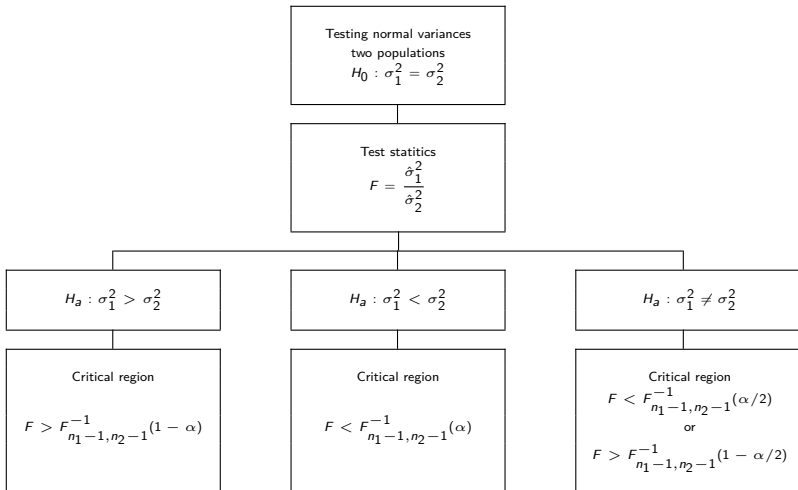
Since Σ_n does not exceed $\chi_{n-1}^{2-1}(1-\alpha)$, we do not reject H_0 .

Conclusion:

➡ There is not sufficient evidence to stop the production process.

Comparing two gaussian populations

The test of the means depends on the test of the variances.



Exercise

Two different scales are to be compared as to the variability they have for weights around 5.0 grams. One object is weighed 11 times on scale 1, and the computed standard deviation is 3.4 grams; the object is weighed 21 times on scale 2 and yields a standard deviation of 5.2.

Based on this information, can we conclude that the scales have different variabilities of measurement? (Use $\alpha = 0.1$)

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Based on this information, can we conclude that the scales have different variabilities of measurement? (Use $\alpha = 0.1$)

Solution:

$$H_0 : \sigma_1 = \sigma_2 \quad \text{v.} \quad H_0 : \sigma_1 \neq \sigma_2.$$

$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} = \frac{(3.4)^2}{(5.2)^2} = 0.4275.$$

$$n_1 - 1 = 10, \quad n_2 - 1 = 20, \quad \alpha = 0.1$$

$$F_{n_1-1, n_2-1}^{-1}(1 - \alpha/2) = F_{10, 20}^{-1}(0.95) = 2.348 \quad (\text{qf}(0.95, 10, 20)).$$

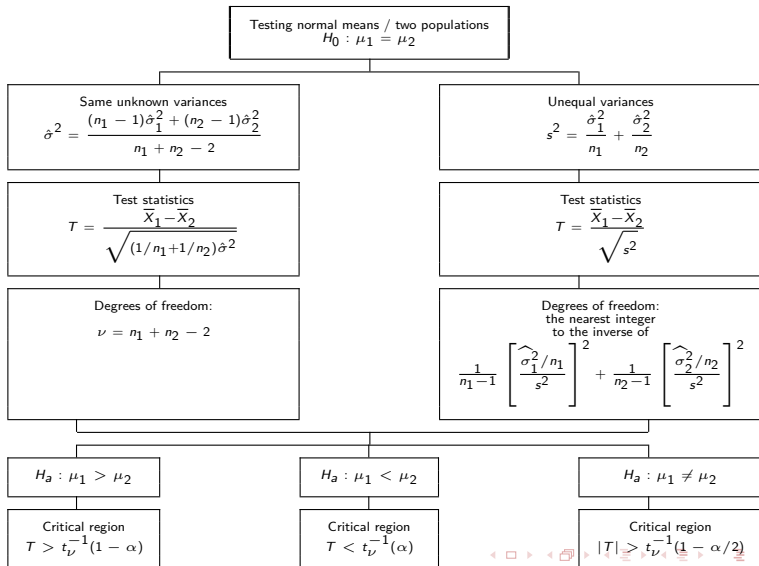
$$F_{n_1-1, n_2-1}^{-1}(\alpha/2) = F_{10, 20}^{-1}(0.05) = 0.360 \quad (\text{qf}(0.05, 10, 20)).$$

Since $F = 0.4275 > 0.360$, we do not reject H_0 .

Conclusion: They experience does not allow to consider that the precision of the scales are different.

➡ The observed difference is due solely to chance.

Testing normal means (two populations)



Testing normal means (two populations)

Exercise

Consider the previous exercise with the two scales. Suppose that in the sample of size 11 we found $\bar{X}_1 = 4.82$ grams, and in the sample of size 21 we found $\bar{X}_2 = 5.81$ grams.

Are the scale reporting the same mean weight on the average?

Testing normal means (two populations)

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Consider the previous exercise with the two scales. Suppose that in the sample of size 11 we found $\bar{X}_1 = 4.82$ grams, and in the sample of size 21 we found $\bar{X}_2 = 5.81$ grams.

Are the scale reporting the same mean weight on the average?

Solution:

$$H_0 : \mu_1 = \mu_2 \quad \text{v.} \quad H_a : \mu_1 \neq \mu_2.$$

$$\widehat{\sigma}^2 = 21.88 \quad \text{and} \quad T = -0.5686.$$

$$t_{n-2}^{-1}(1 - \alpha/2) = t_{30}^{-1}(0.975) = 2.042$$

Since $|T| = 0.5686$ is not larger than $t_{n-2}^{-1}(\alpha/2) = 2.042$, we do not reject H_0 .

Conclusion: We cannot say that the two scales do not have the same accuracy ($\mu_1 = \mu_2$) and the same precision ($\sigma_1^2 = \sigma_2^2$).

➡ The observed differences are due to random variability solely.

In practise, we will consider that the two scales have the same accuracy and the same precision, that they are identical.